# A35: The space spanned by the 35-component law vector, made exact

**For:** Arvin Hampton (539 Labs) · 2026-10-22, about 10:55 AM CDT.

**Folder:** `reports/s_family_paths/A35_lawspace_exact/`

- Scripts: `symbols.py`, `orbits.py`, `exact_dim.py`, `exact_main.py`, `exact_intertwiner.py`, `floatcheck.py`.
- Outputs: `exact_main.json`, `exact_dim.json`, `exact_intertwiner.json`, `quot_*.json`, `X_modp_*.npy`.

**Ground rules**

- CORE_FREEZE: H_s was not redefined and certifier code was not touched.
- Read-only imports: A32 (`exact_hom.py`, `exact_verify.json`), `A/scripts/weil_family.py`, and A33's `law.py`. The last is used only in the floating-point sanity check.
- New files only. Nothing was pushed or sent. No TPC or RH claims.
- **Cat A** = proved, or a computer certificate (CC) with exact arithmetic. **Cat B** = floating point.

**Notation**

- The **35-component law vector** has components $G_{\alpha,\beta}(x)$ = mean over $\nu \in \alpha+Z^2$ of $e(x\cdot Q(\nu) + B(\nu,\beta))$, for $x \in Q$, with $A = [[2,7],[7,7]]$.
- Its characteristic classes are, for $\mu \in D = A^{-1}Z^2/Z^2$:
- $K2_\mu = G_{a+\mu,0}$
- $K3_\mu = G_{\mu,a}$
- $K4_\mu = G_{\mu,0}$
- $K1_\mu = G_{a+\mu,a}$

Here $a = (7/10, -1/5)$ and $h_A = \frac{1}{2}A^{-1}A^* = (1/2, 0)$.

- **V** is the space of functions on Q spanned by $K2 \cup K3 \cup K4$.

# Summary

| Item | Status |
|---|---|
| (1) Profile S-law and T-law, closed-form $M_S$ and $M_T$ | **Proved** (Cat A). Correction: the factor is $\mathbf{\lvert x\rvert^{-1}}$, not $x^{-1}$ |
| (2) Exact basis, dim V = 36, closure of V under S and T | **Proved / certified** (Cat A; upper bound proved, lower bound CC at two primes) |
| (3) $V \cong E$ with an explicit cyclotomic intertwiner; $\mathrm{Hom}(V\otimes\chi^k, C) = 0$ for all k | **Proved** (Cat A; intertwiner verified exactly in $Q(\zeta_{58800})$; Hom vanishing CC at two primes) |
| (4) $K1 \equiv 0$ | **Proved** (Cat A) |

# 1. The laws (Cat A, proof)

**Elementary rules.** These hold for all $\alpha, \beta \in Q^2$ and all x:

- $G_{\alpha+n,\beta} = G_{\alpha,\beta}$ for $n \in Z^2$.
- $G_{\alpha,\beta+\lambda} = e(B(\alpha,\lambda))\cdot G_{\alpha,\beta}$ whenever $A\lambda \in Z^2$. The reason is that $B(n,\lambda) = n^T A\lambda \in Z$.
- $G_{g\alpha,g\beta} = G_{\alpha,\beta}$ for every automorph $g \in GL_2(Z)$ with $g^T A g = A$. This is a change of

variables $\nu \mapsto g\nu$ on the lattice coset; the integrand is periodic, so the mean is preserved.

**T-law.** For all $\alpha$, $\beta$ and all $x$:

$$G_{\alpha,\beta}(x+1) = e(-Q(\alpha) - B(\alpha,h_A))\cdot G_{\alpha, \beta+\alpha+h_A}(x).$$

*Proof.* Write $\nu = \alpha+n$. Then $Q(n) = n_1^2 + 7n_1n_2 + (7/2)n_2^2 \equiv (7/2)n_2 \equiv \tfrac{1}{2}n^TA^* = B(n,h_A) \bmod 1$, using $n_2^2 \equiv n_2 \bmod 2$. Expanding gives $Q(\nu) \equiv B(\nu, \alpha+h_A) - Q(\alpha) - B(\alpha,h_A) \bmod 1$, which is the law.

**S-law.** For all $\alpha$, $\beta$ and all $x \in Q$ with $x \neq 0$:

$$G_{\alpha,\beta}(-1/x) = (1/(|x|\cdot\sqrt{35}))\cdot e(B(\alpha,\beta))\cdot \Sigma_{p\in D}\, G_{\beta+p,-\alpha}(x).$$

*Proof (Poisson summation).*

1. **Set-up.** Put $Z = yA + itI$ with $t > 0$, and $f(\nu) = \exp(\pi i\cdot \nu^T Z\nu + 2\pi i\cdot B(\nu,\beta))$.
2. **Poisson summation:** $\Sigma_{\nu\in\alpha+Z^2} f(\nu) = \det(-iZ)^{-1/2} \cdot \Sigma_{w\in Z^2} e(w^T\alpha) \cdot \exp(-\pi i\cdot (w-A\beta)^T Z^{-1}(w-A\beta))$.
3. **Substitution.** Put $w - A\beta = -A\gamma$. Then $\gamma$ runs over $\beta + A^{-1}Z^2$, and $e(w^T\alpha) = e(B(\alpha,\beta) - B(\gamma,\alpha))$.
4. **Left side as t → 0.** The integrand is periodic, so a finite Fourier expansion and Poisson summation for the Gaussian give $t\cdot\mathrm{LHS} \to G_{\alpha,\beta}(y)$, with an exponentially small error.
5. **Right side.** - **Determinant.** $A$ has eigenvalues $\lambda_1 > 0 > \lambda_2$. With principal branches, $\det(-iZ)^{1/2} \to \sqrt{(|y|\lambda_1)}\cdot e^{\mp i\pi/4} \cdot \sqrt{(|y||\lambda_2|)}\cdot e^{\pm i\pi/4} = |y|\cdot\sqrt{35}$, for either sign of $y$. Signature 0 means no eighth root of unity survives. - **Expansion of the quadratic form.** $A\cdot Z^{-1}\cdot A = A/y - (it/y^2)\cdot I + O(t^2)$. The $O(t^2)$ term changes the phase by $O(t^2|\gamma|^2)$, which contributes $O(1)$ against a main term of order $1/t$. - **Mean on the dual side.** The main term is a periodic phase $e(-Q(\gamma)/y - B(\gamma,\alpha))$ on the coset $\beta + A^{-1}Z^2$ (density 35) times the Gaussian $e^{-\pi t|\gamma|^2/y^2}$. Its sum is $35\cdot(\text{mean})\cdot y^2/t$, and the mean over $\beta + A^{-1}Z^2$ equals $(1/35)\cdot\Sigma_p\, G_{\beta+p,-\alpha}(-1/y)$.
6. **Conclusion.** Together: $G_{\alpha,\beta}(y) = (|y|/\sqrt{35})\cdot e(B(\alpha,\beta))\cdot\Sigma_p\, G_{\beta+p,-\alpha}(-1/y)$. Putting $y = -1/x$ gives the stated law. ∎

**Closed forms (Cat A).** `symbols.py` reduces every term with the elementary rules. On the 105 symbols K2∪K3∪K4:

- **M_S:** each row is $(1/\sqrt{35})\cdot\Sigma$ of 35 terms of the form $e(k/9800)$.
- **M_T:** monomial. It acts diagonally on K2 and swaps K3_μ ↔ K4_μ, with $e(k/9800)$ phases.

The family K1 is closed on its own under both laws.

**Sign of x (Cat B corroboration, `floatcheck.py`).**

- On 72 rationals of both signs: the $|x|^{-1}$ law has error $5\cdot10^{-15}$ and the T-law $1\cdot10^{-15}$.
- The signed $x^{-1}$ version **fails for x < 0** (error 2).
- So item (1) as worded ("$x^{-1}$") is correct only for $x > 0$. The correct weight factor is $|x|^{-1}$.

# 2. Exact basis and dim V = 36 (Cat A)

- **Upper bound (proved).**
- The automorph group search (`orbits.py`) gives 12 elements with $|\text{entries}| \le 40$, including $\pm I$ and $\gamma = [[-1,-7],[2,13]]$.
- Under these elements, the 105 symbols fall into **36 orbits**: 12 in each of K2, K3 and K4, of sizes 1, 2 and 4.
- Within an orbit, every symbol equals $e(k/9800)$ times the orbit representative. No orbit is forced to vanish.
- So V is spanned by the 36 orbit representatives, and **dim V ≤ 36**.
- Cross-check: the relation rank is 69 at two primes, and the relation span is closed under S and T.
- **Lower bound (CC).**
- The Gauss means of all 105 symbols were evaluated **exactly mod p** at the 48 rationals $h/m$ with $m \le 6$ and $-2m \le h < 2m$.
- The primes were $p = 1\,764\,001$ and $6\,468\,001$, both $\equiv 1 \bmod 588\,000$, with $\zeta$ mapped to a

root of unity in F_p.
- The rank is **36** at both primes.
- The values lie in Z[ζ][1/(4m)²], so the rank can only drop under reduction. Hence the char-0 rank is ≥ 36, and **dim V = 36**.
- **Consequences.**
- The 36 orbit representatives form an **exact basis** of V.
- The relation space is exactly the kernel of evaluation.
- **V is closed under S and T**, because both laws hold for all x and map the symbols into the symbols.
- The quotient matrices M_S and M_T on V are explicit cyclotomic matrices. They satisfy $S^4 = 1$ and $(ST)^3 = S^2$ mod p, for exactly one of the two roots of 35 mod p. That root is the same one that ρ_7|W selects.

# 3. V ≅ E, and no $\chi^k$-twisted map into C (Cat A)

**Mod-p Hom dimensions** (`exact_main.json`; A32's method; identical at both primes):

- End(V) = 1.
- Hom(V, E) = Hom(E, V) = Hom(V, W) = 1.
- **Hom(V⊗$\chi^k$, C) = Hom(C, V⊗$\chi^k$) = 0 for all k = 0, …, 11.**
- Hom(V⊗$\chi^k$, E) = 0 for k ≠ 0.
- Each mod-p dimension is an upper bound for the char-0 dimension, so every vanishing above is **exact**.
- E and C are coordinate blocks of W (orbit indices 26, 28, …, 36 form C), and the block-diagonal form was verified exactly.

**Explicit intertwiner** (`exact_intertwiner.py`), with N = 58 800:

- **T-eigenbases.** On V, T is monomial with 1- and 2-cycles, and its T-eigenbasis is exact over Z[½, ζ_N]. On E, T is likewise monomial.
- **T-spectra.** Both spectra consist of the same **36 distinct** values, in exact agreement.
- **The map.** X = P_E · Π · diag(c) · $P_V^{-1}$, where:
- P_V and P_E are the exact eigenvector matrices ($P^{-1}P = I$ checked exactly);
- Π matches equal eigenvalues;
- **every c_i is a root of unity ζ_N^{e_i}**. The exponents are listed in `exact_intertwiner.json`.
- **Exact verification.**
- X·S_V = S_E·X: **0 of 1296 entries fail**, checked by reduction modulo the cyclotomic polynomial $\Phi_{58800}$. The factor 1/√35 (positive root on both sides) cancels.
- X·T_V = T_E·X holds by the exact eigenvalue match.
- X is invertible because it is a product of invertible factors.
- **Therefore V ≅ E, with no twist (proved).**
- Since dim V = dim E = 36, together with the vanishing Hom(V⊗$\chi^k$, C) = 0 for every k: **the 35-component law vector has no image in C under any η^{2k} twist, including $\chi^3$ ($\eta^6$).** This upgrades A33's Cat B statements to Cat A.

# 4. K1 ≡ 0 (Cat A, proof)

1. **Zero at x = 0.** G_{α,a}(0) = e(B(α,a))·mean over n ∈ $Z^2$ of $e(n^T Aa)$. Here Aa = (0, 7/2), so the mean is the mean of $(-1)^{n_2}$, which is 0. So K1(0) = 0.
2. **Closure.** The family K1 is closed under both laws (`symbols.py`): T maps K1 to K1 monomially, and S maps K1 to combinations of K1.
3. **Induction on the denominator.** - For an integer x = n: K1(n) = $M_T^n$·K1(0) = 0. - For x = h/m with m ≥ 2: write x = n + r/m with 0 < r < m. By the S-law, K1(r/m) = (r/m)·M_S·K1(−m/r), and −m/r has the smaller denominator r.
4. **Conclusion.** Hence **K1 ≡ 0 on Q**, matching A33's floating-point observation.

# Not claimed

- **Scope of the S-law.** It is proved for the leading radial profiles (the 35-component law vector) only. Nothing is claimed about the cone sums beyond leading order.
- **Gauss-mean values.** The evaluations used for the lower bound are mod-p certificates, not

full cyclotomic values.

# Status

- **Cat A:** items (1)–(4) as proved or certified above. The only correction is the $|x|^{-1}$ weight factor.
- **Cat B:** only the floating-point corroboration in `floatcheck.py`.